



FINDING THE SHORTEST PATH
IN
AGENT BASE MODELS
PROJECT
BASED ON VISUAL ACUITY (VA), INDOORS

BOOTCAMP
2024



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Morphotect



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SUPPORT TEAM



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GRAPHIC DESIGNER
RESERCHER



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METADATA

OUR TEAM

ABOUT PROJECT WHO I AM ?

I AM A HUMAN BEING.

The total of variables and data .

I function in the form of a complex nervous system and live in my earthly body.

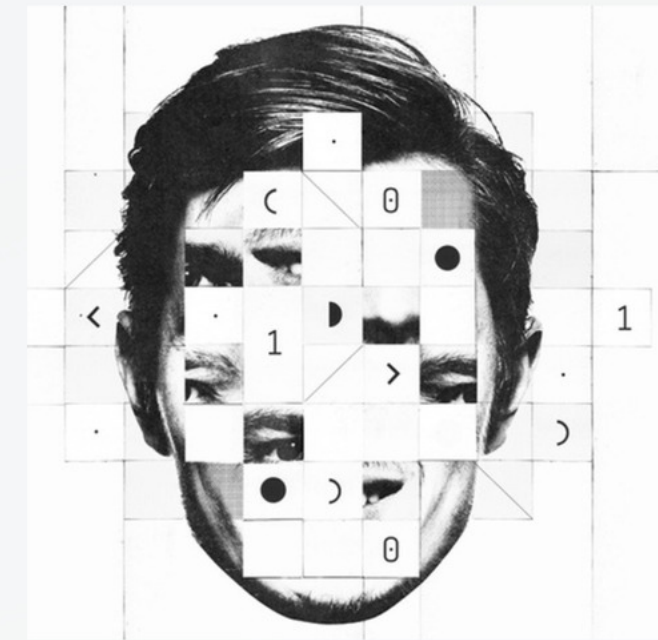
I AM A COLLECTION OF SENSES AND PERCEPTION.

I have five main senses naturally, the most important of which is **vision**.

“The sense of sight as the most universal and original feeling among the five human senses.”
(Descartes)

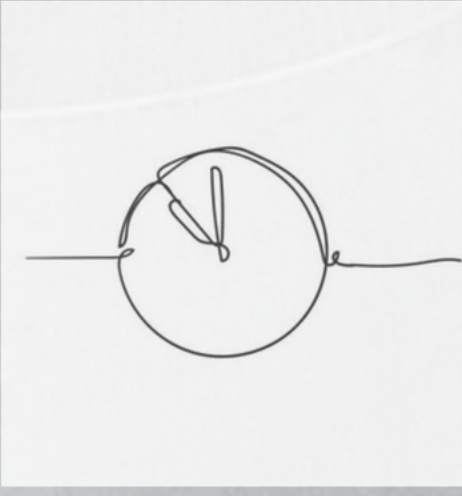
“In real-life experiences, the senses of sight and touch are closely related.”
(Herbert Bayer, The Lonely Metropolis, 1932)

HUMAN EXPERIENCES ARE A COLLECTION OF PICTURE FRAMES.



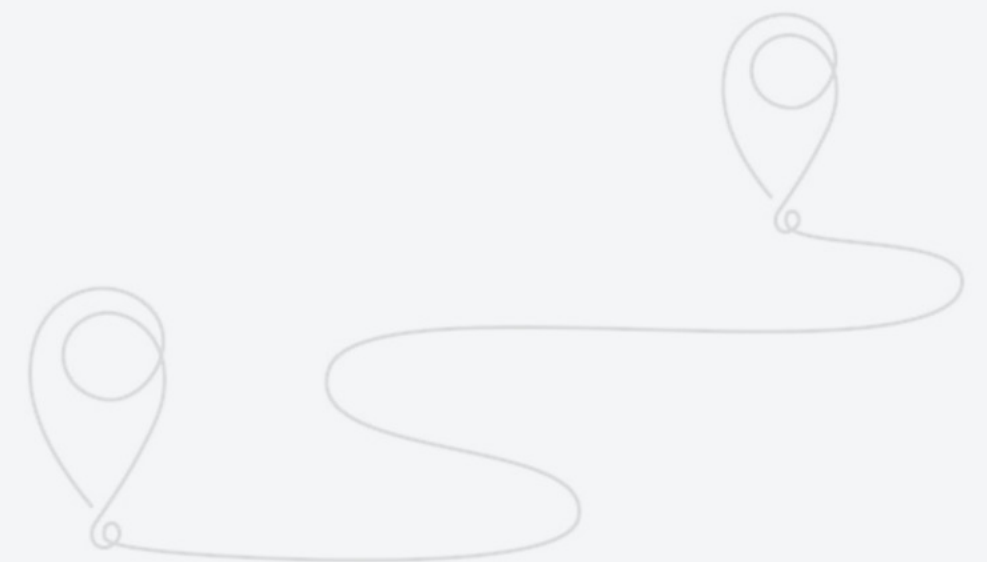
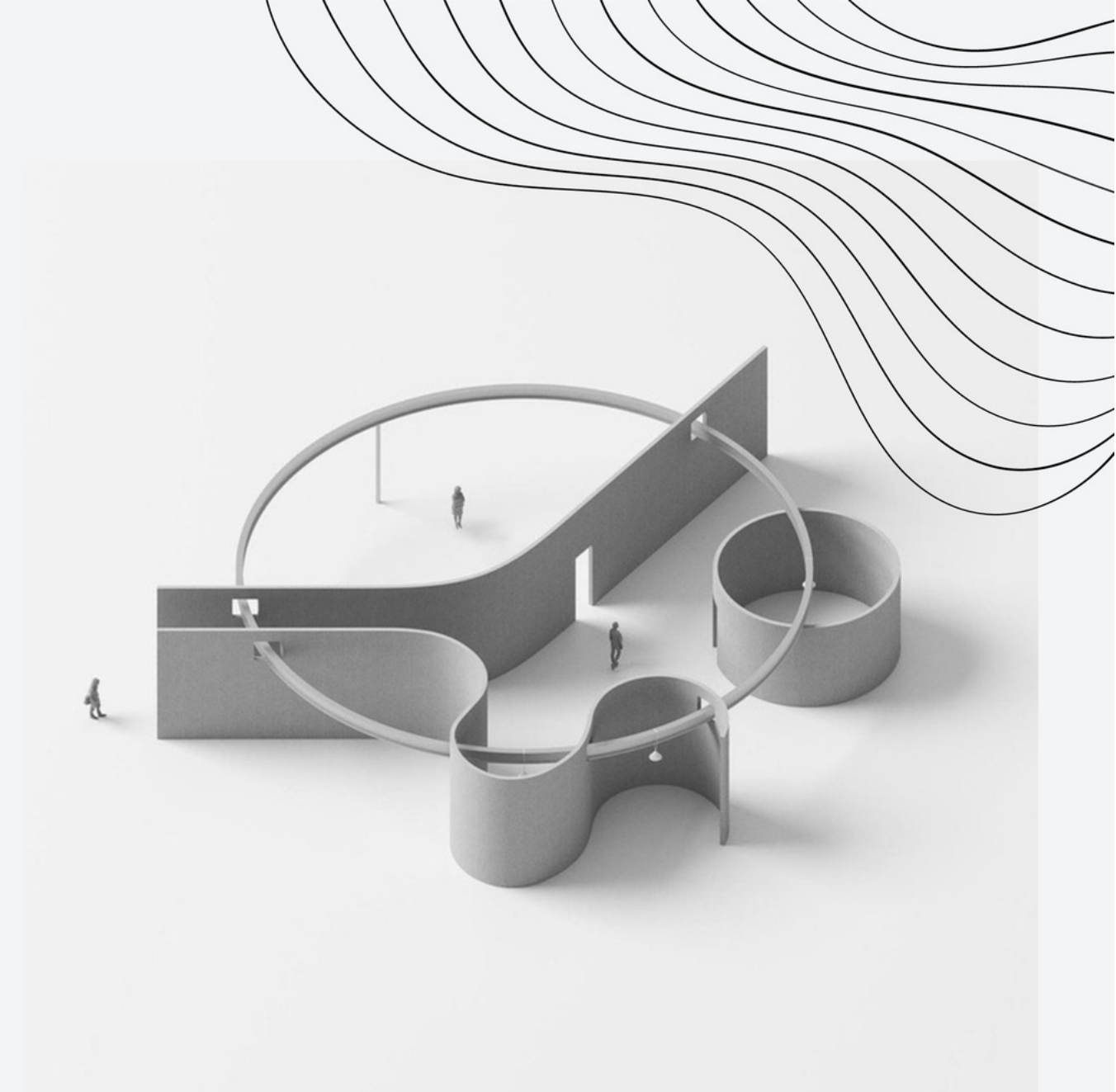
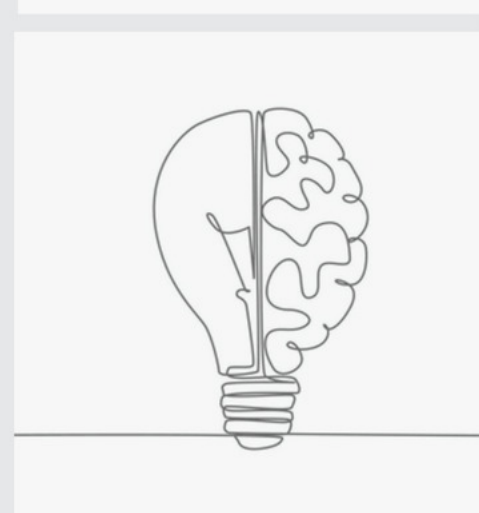
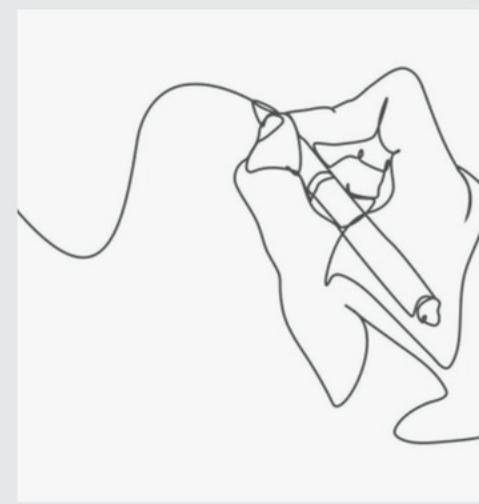
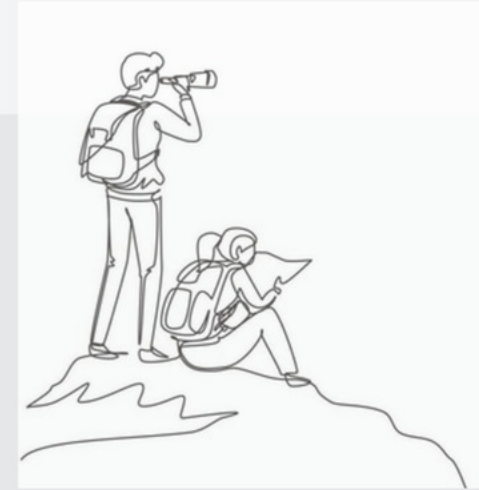
CONCEPT IN PROJECT

Faced with an unknown place, every person naturally seeks to find the most suitable path to reach the desired place. Optimum routing is done by combining the movement of the nervous system and the five human senses, specifically, the sense of sight, as well as the assessment of the user's vision limit.

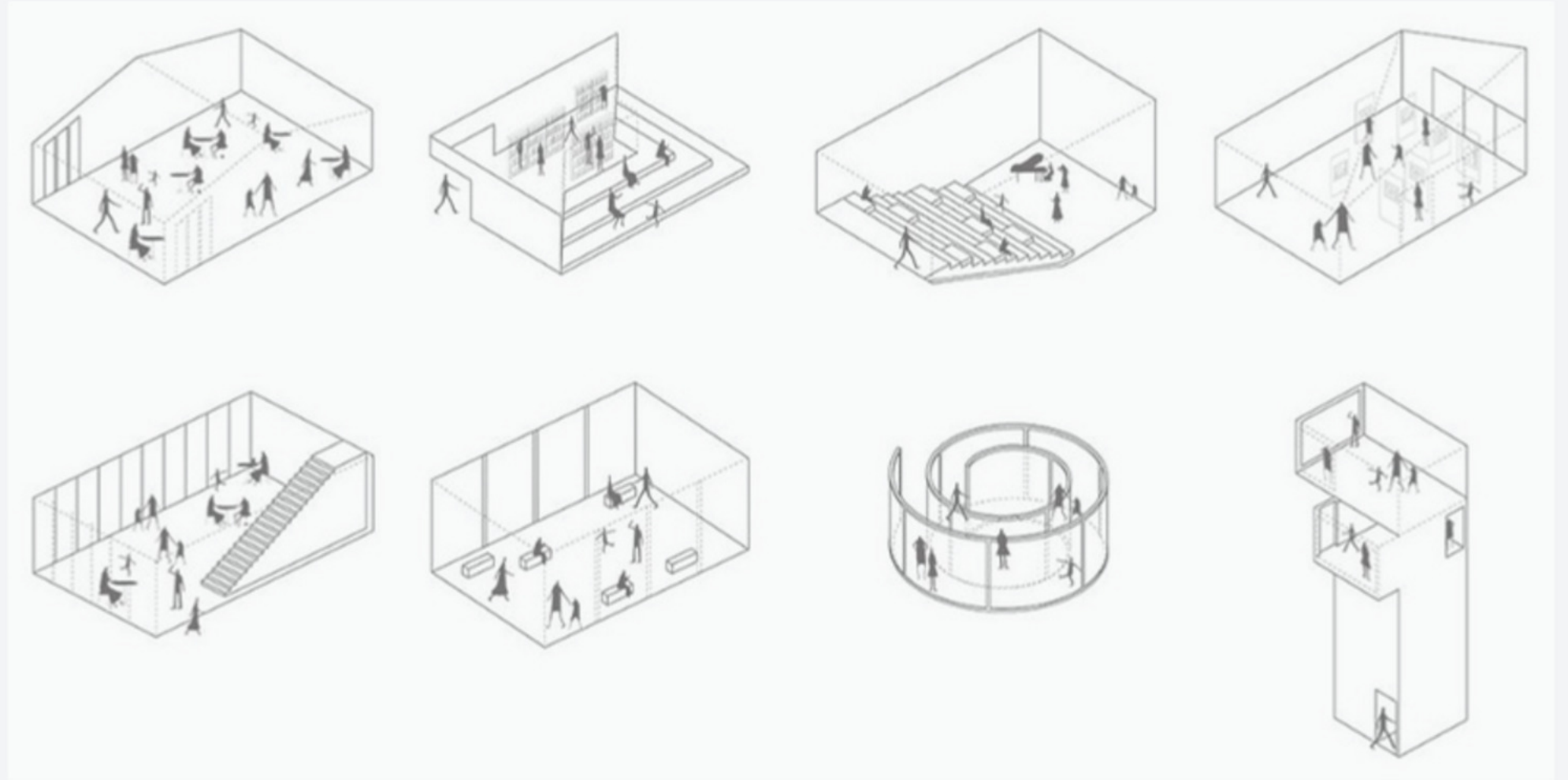


STRATEGIES

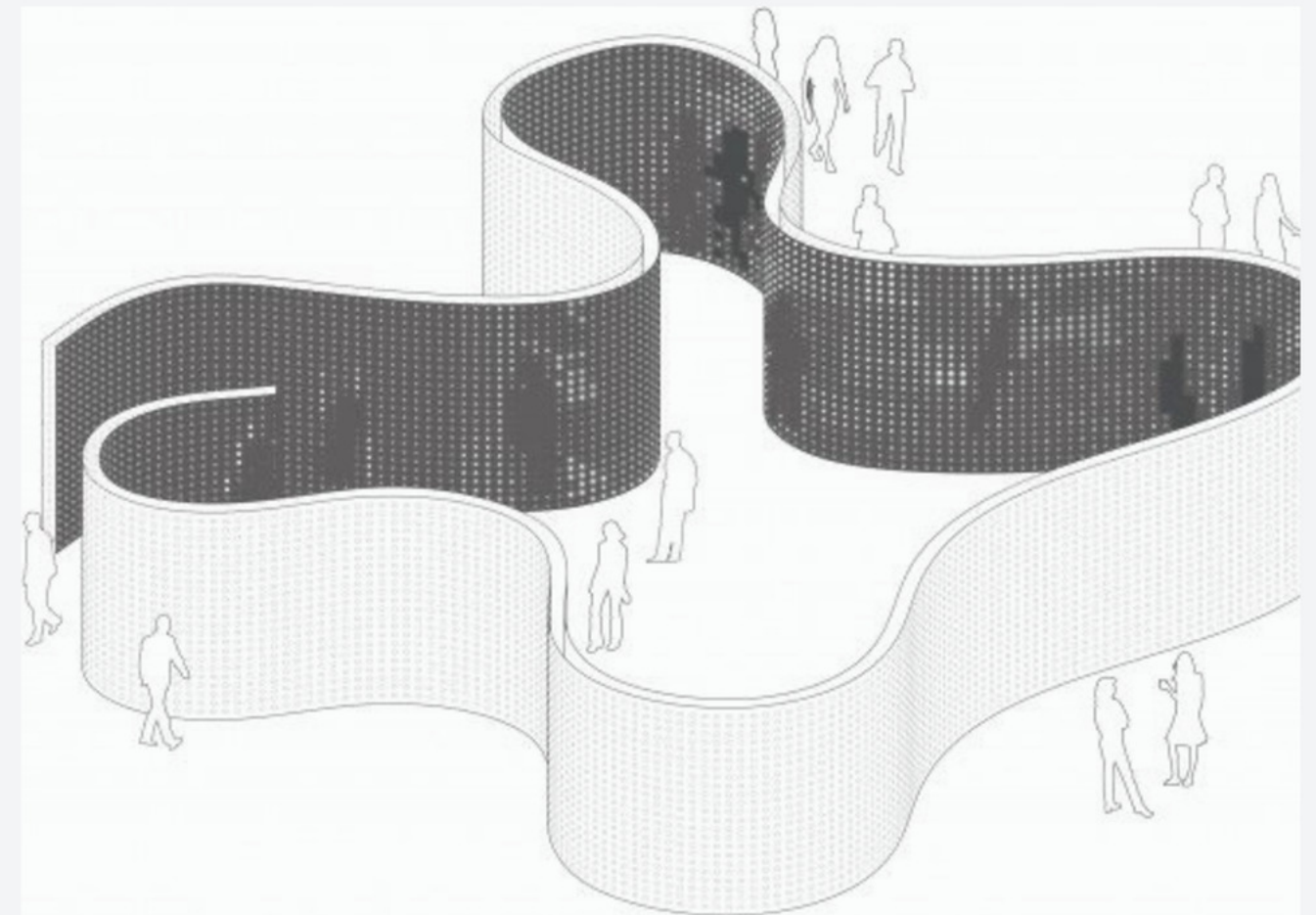
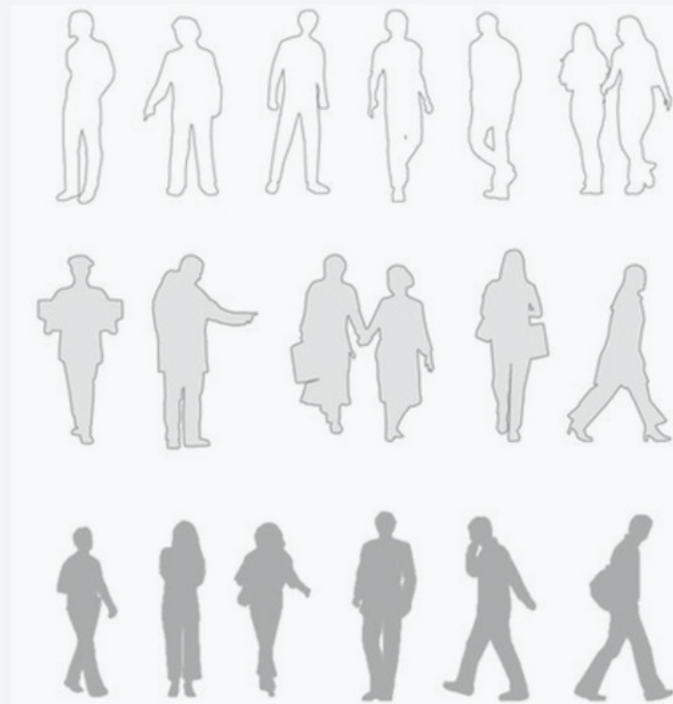
Legibility is a quality that makes a place comprehensible, and it finds meaning in two levels of physical form and activity patterns. To take full advantage of the potential facilities of a place, awareness of physical form and activity patterns should complement each other. This issue is especially important for foreigners, because they need to understand the place quickly without any previous preparations (Bentley et al., 1985). Legibility is a quality of the environment that makes a person remember his surroundings well and prevents the user from getting confused in the environment (Beiniaz and Hanai; 2016), also the legibility of the space is the user's need for safety, aesthetics and navigation. (Eraydin, 2007)



VARIABLES



VISION & PATH



SOLUTION FINDING PATH BY ALGORITHM A*

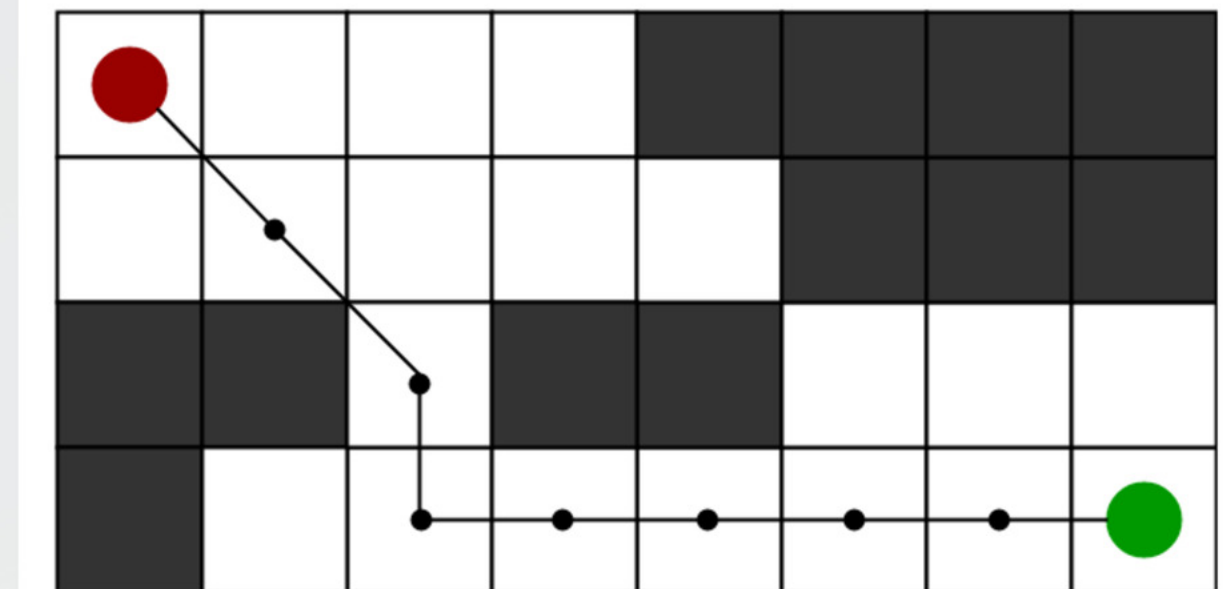
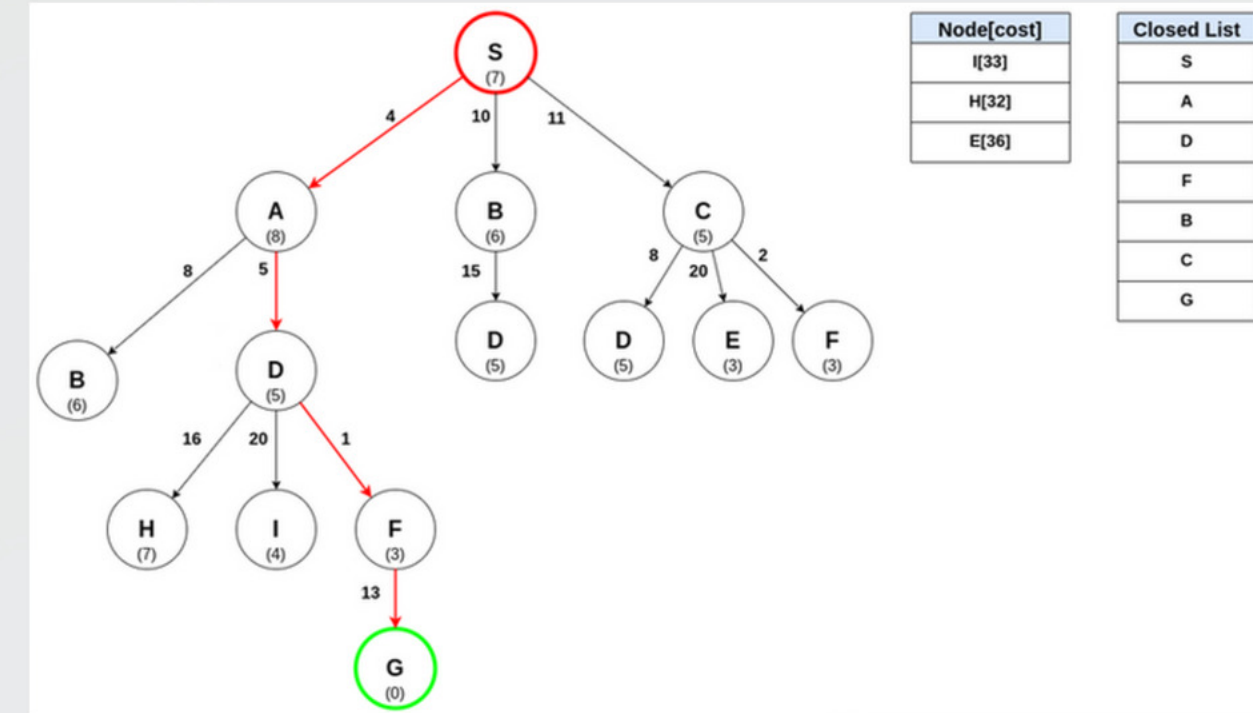
Which way is better?

THIS IS PATH ☒

THIS IS PATH 

THIS IS PATH ☒

THIS IS PATH 



SOLUTION

Visual Acuity

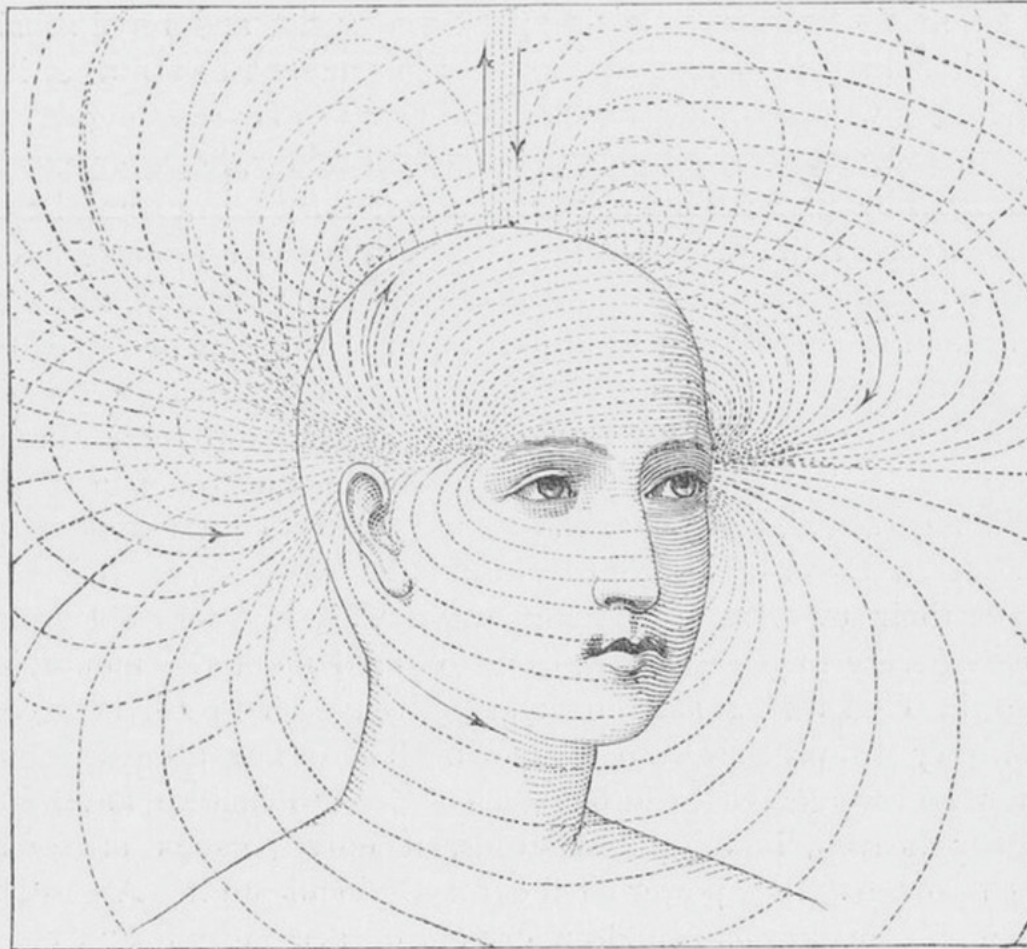
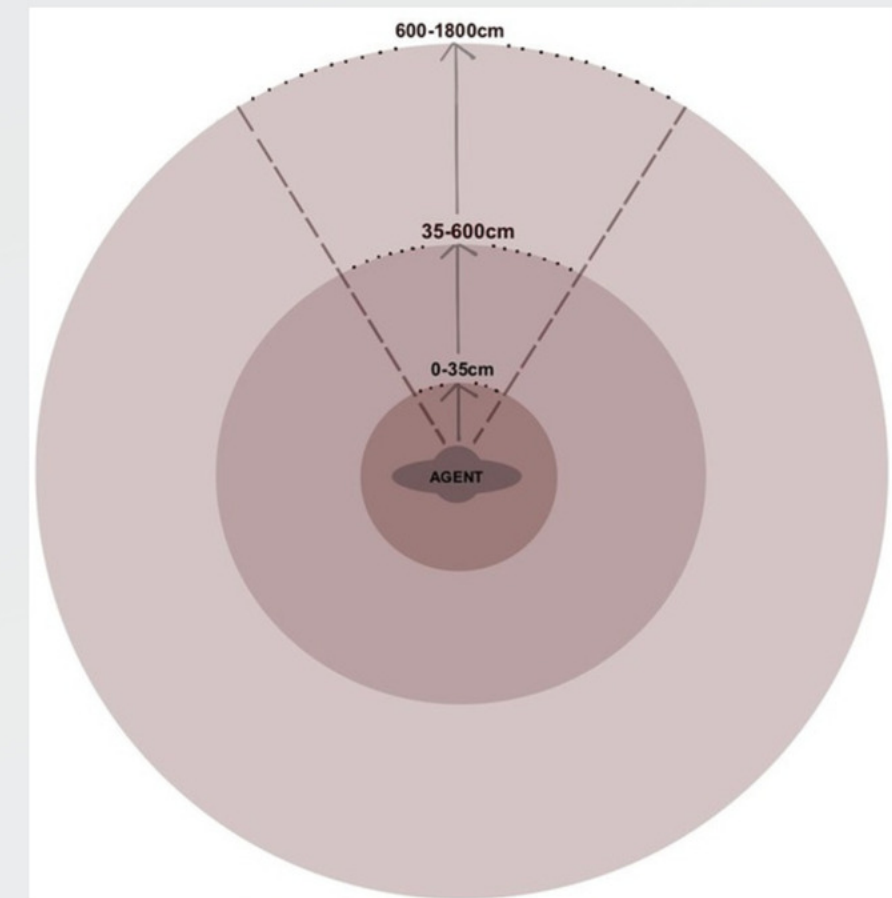
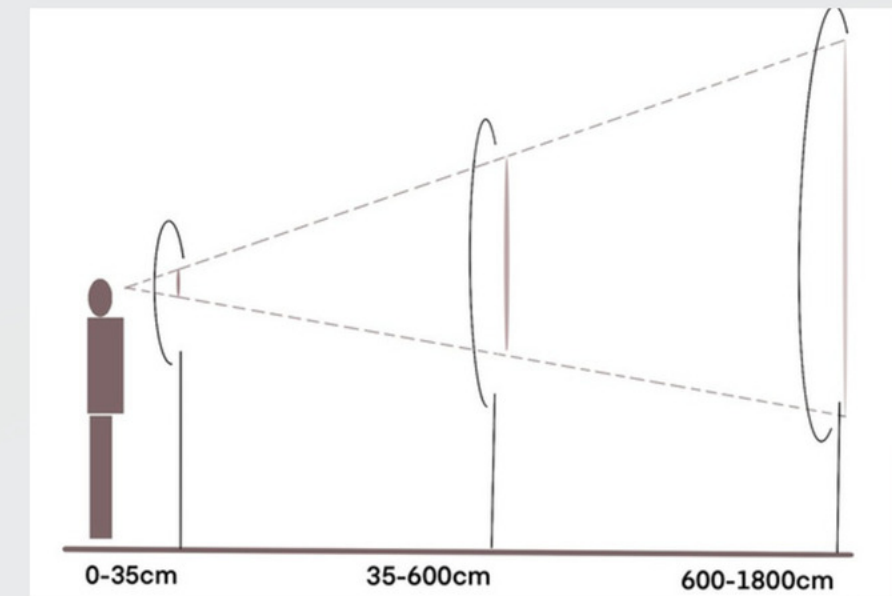
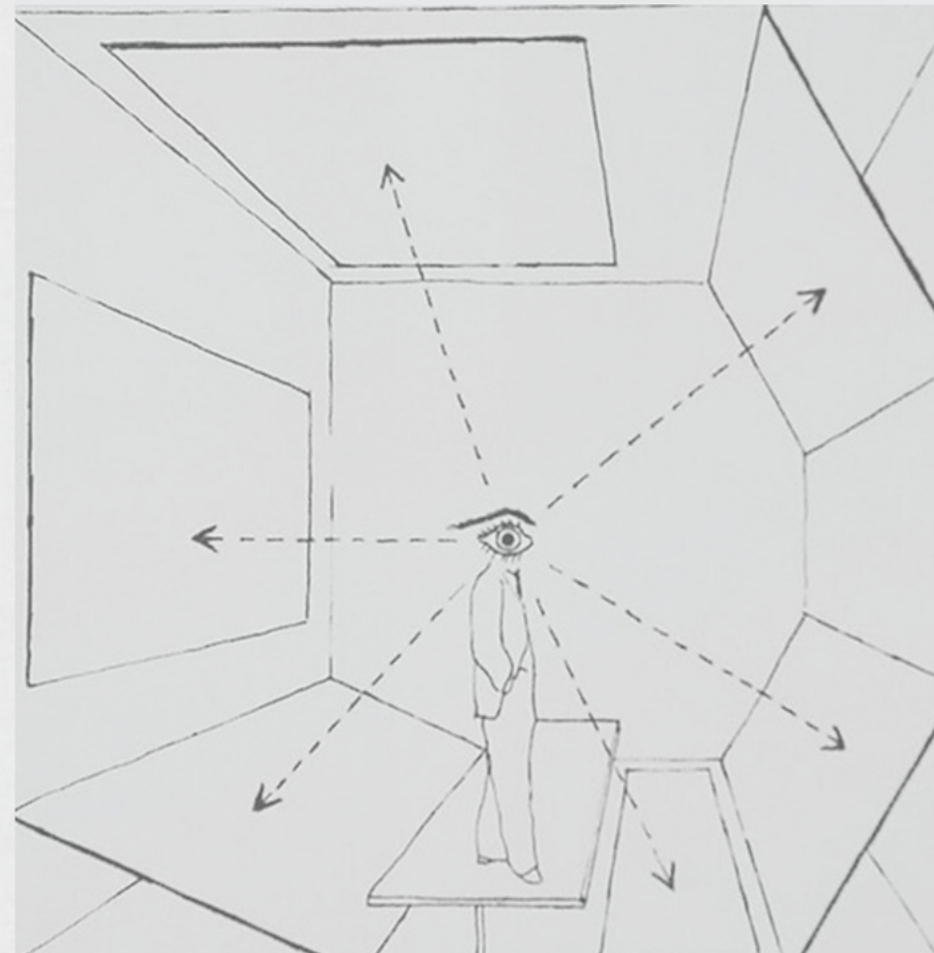


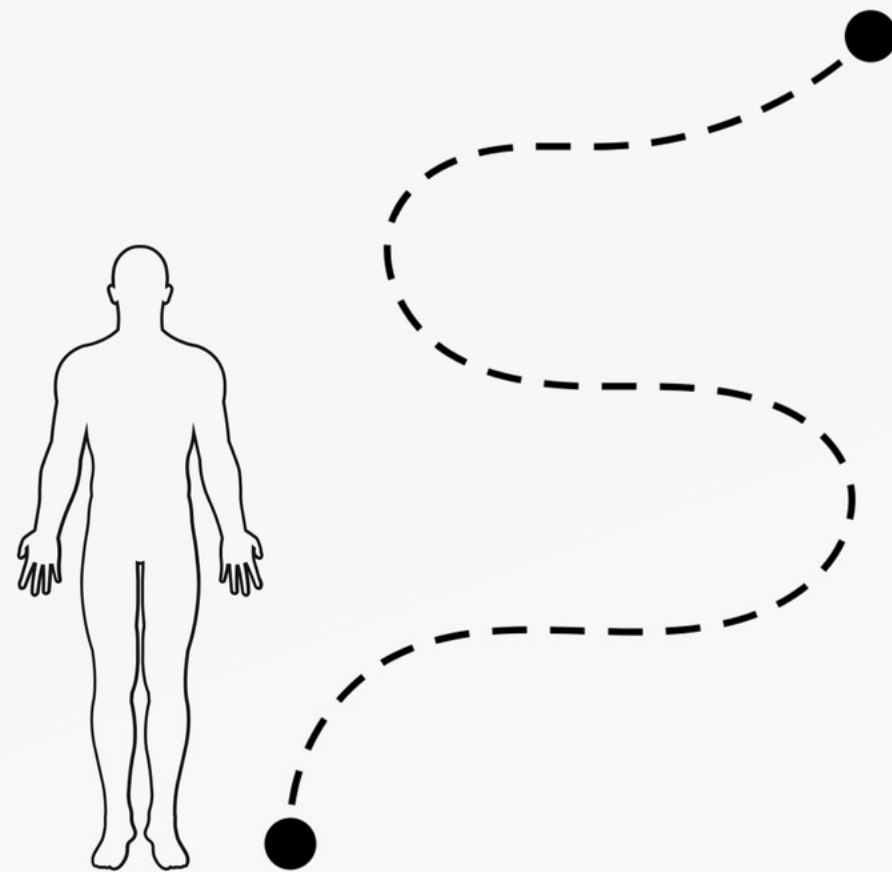
Fig. 187. The Psycho-Magnetic Curves.

In medical sciences, visual acuity can be evaluated and measured by displaying targets of different sizes that are shown at a standard distance from the eye. Generally, in medicine, human vision is measured at a distance of 6 meters (far vision) or at a distance of 35 cm (near vision), but to measure human reading acuity in the environment and based on the studies conducted, it can be said that that, the human eye has the ability to read texts only in the range above its head at a distance between 0 and 35 cm, at a distance between 35 cm and 6 meters, the best readability and understanding of the surrounding environment is created for the user and From 6 meters to 18 meters, the user can understand the general perception and readability of the surrounding environment (Javadi et al., 2017). The wider the scope of this importance in the spaces outside the building, the more the user's readability of the environment will increase.



THIS PROJECT

In this research, using agent-based models (ABMs), it has been tried to identify the presence of the user in the Faculty of Architecture as one of the high-traffic public spaces that is evaluated under the influence of the user's behavior and knowledge about the surrounding environment. In order to achieve this goal, by examining and analyzing visual acuity and considering three ranges of vision, with the aim of finding the right path for the user to reach the desired destination in the shortest possible time, the behavior of the user in this space has been simulated. .



PSEUDOCODE OF THE PROJECT

1. Import necessary libraries:

- o Rhino.Geometry as rg
- o Math

2. Set domain for the path (the obtained paths from A* algorithm):

- o Create an interval from 0 to 1
- o Assign this interval to the path's domain

3. Initialize or update parameter para:

- o If para is not defined globally or if reset is true, set para to 0
- o Increment para by 0.01

4. Calculate position and tangent b on the path:

- o Get the point at parameter para on the path
- o Get the tangent at parameter para on the path

5. Calculate the normal vector perpendicular to the tangent.

6. Set the Z-coordinate of the position (agent) to 1.8 (the height of the agent).

7. Create a line of sight

- o from the position in the direction of the tangent, scaled by 6 (based on the standard range of the human vision (sight)).
- o Calculate the length of the line of sight.

8. Define a plane at the position with the tangent and normal vectors.

9. Create circles of sight with various radii (specifying the sight and visibility with these sizes):

- o Full length of the line of sight.
- o Half the length of the line of sight.
- o Fixed radius of 0.35. (minimum sight for a human)
- o Fixed radius of 18. (maximum sight for a human)

10. Calculate the direction vector by rotating the tangent vector by 62 degrees. (determine the standard line of sight)

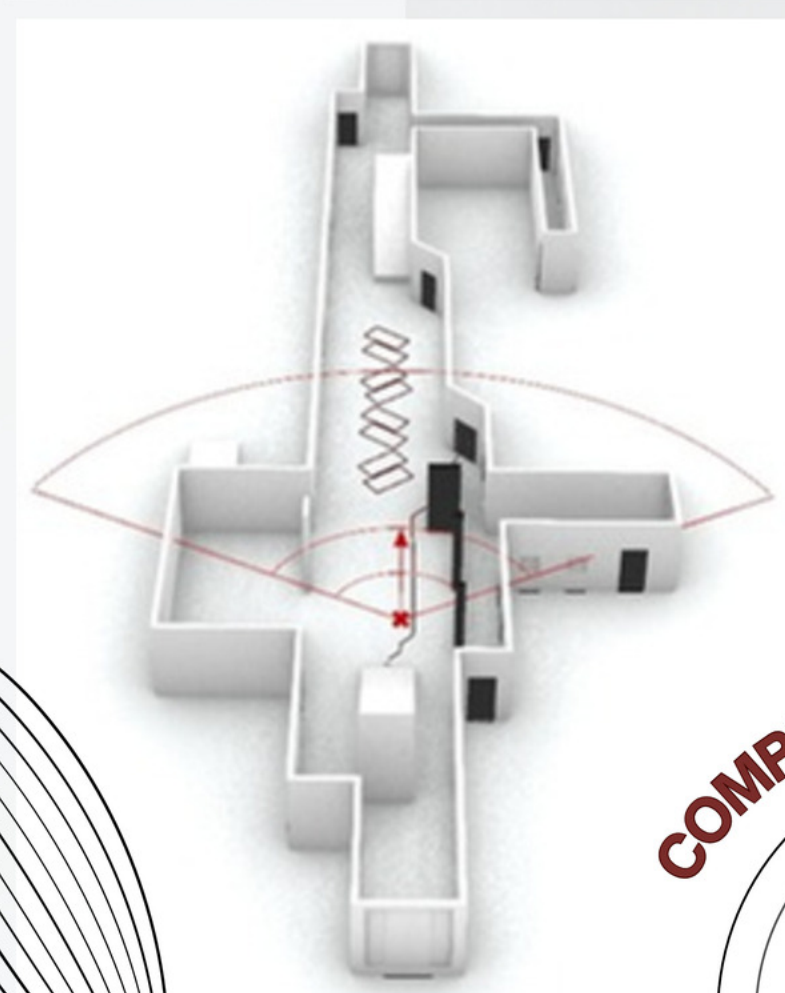
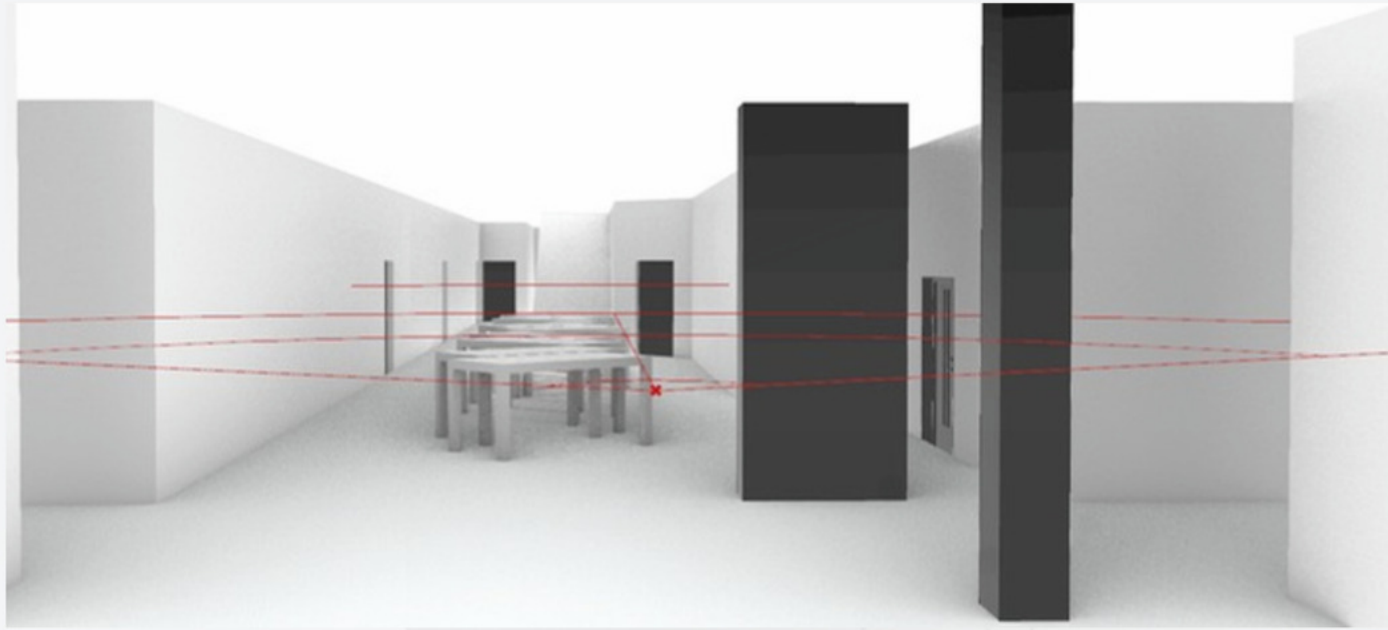
11. Create two new lines from the position in the rotated directions:

- o First line rotated by +62 degrees.
- o Second line rotated by -62 degrees.

12. For each circle of sight (full, middle, shortest, longest):

- o Find intersections between the circle and the two new lines.
- o If intersections are found:
 - Get the intersection points.
 - Find the closest points on the circle to the intersection points.
 - Define an interval between the closest points.
 - Create an arc on the circle within the interval.

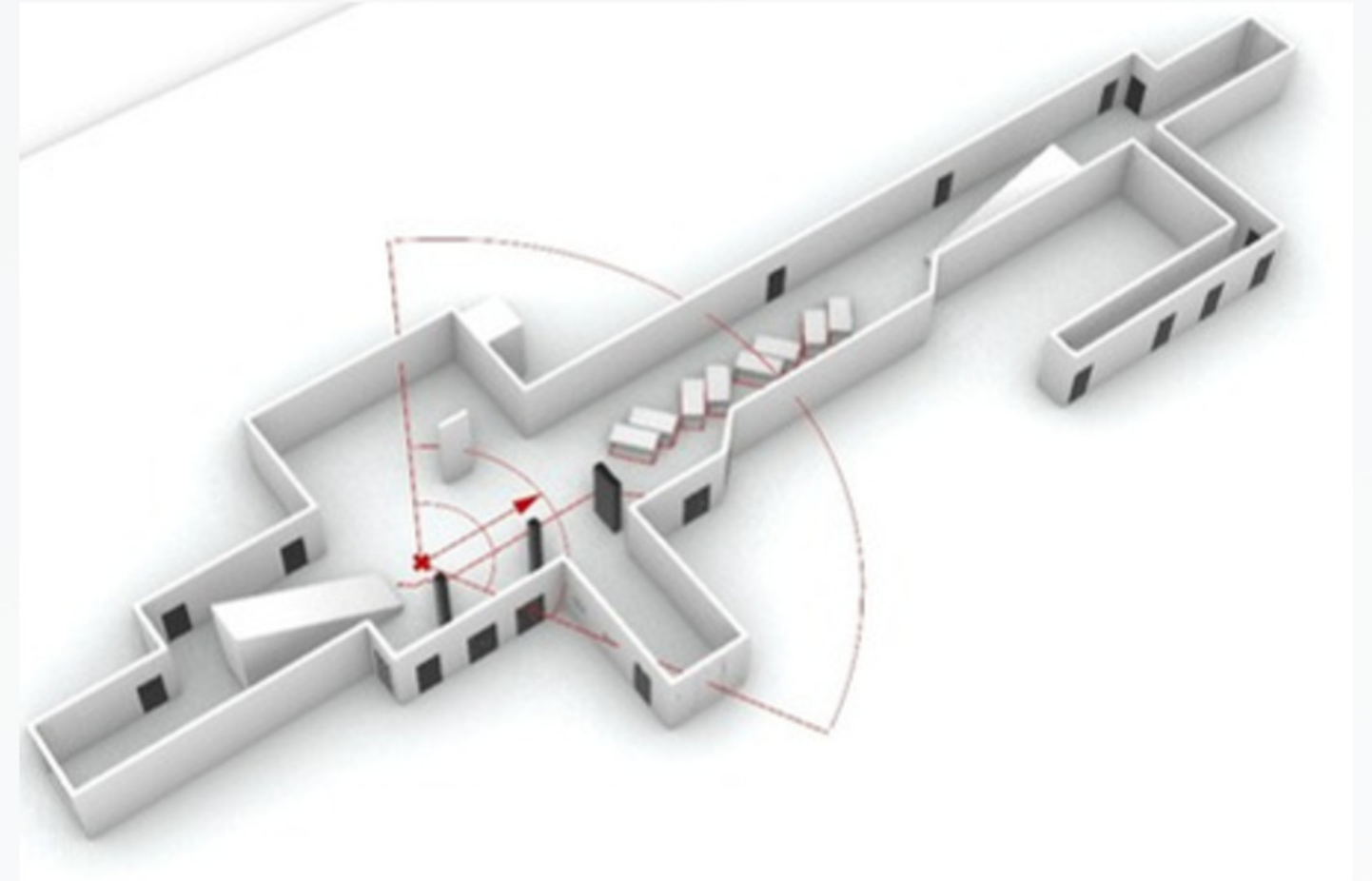
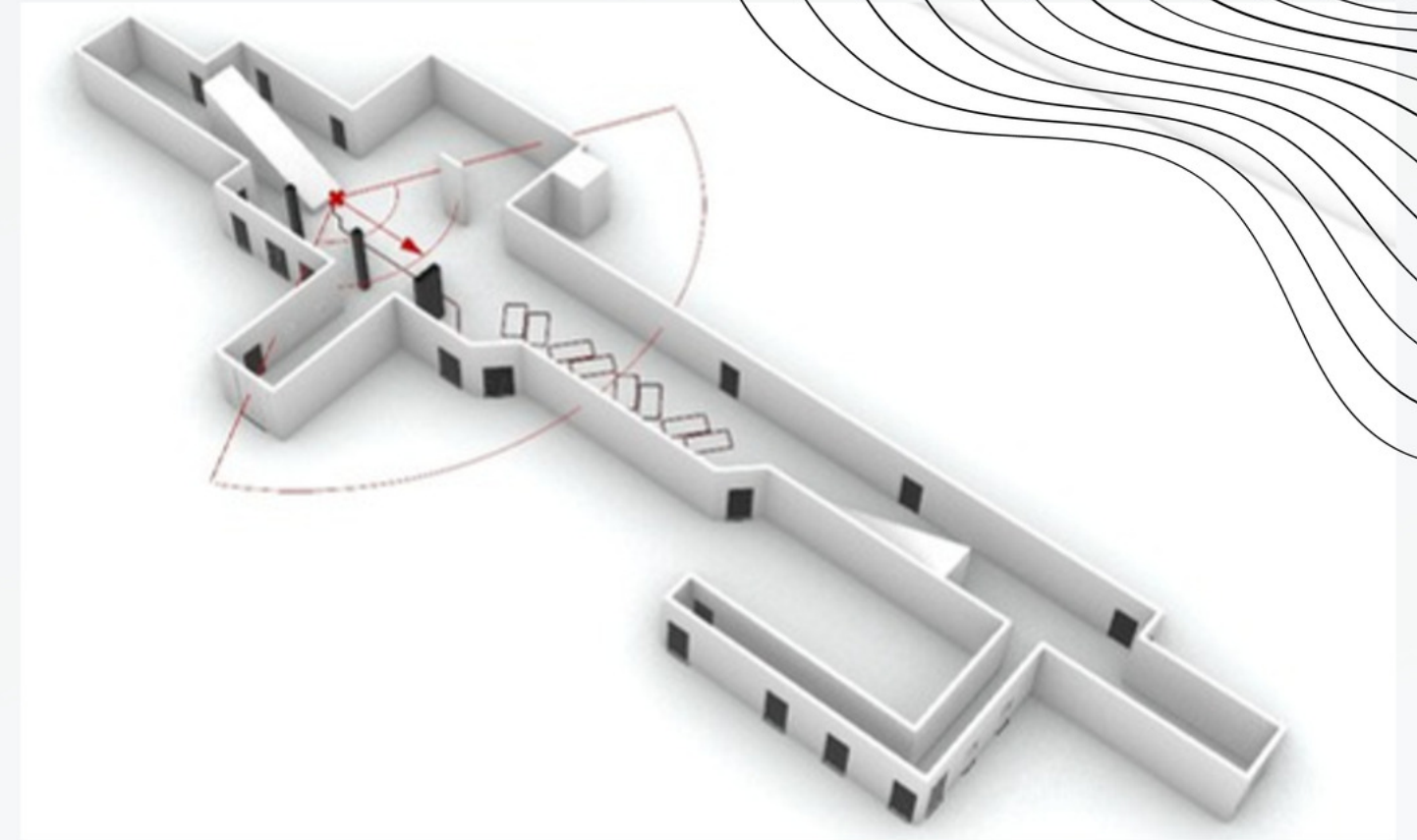
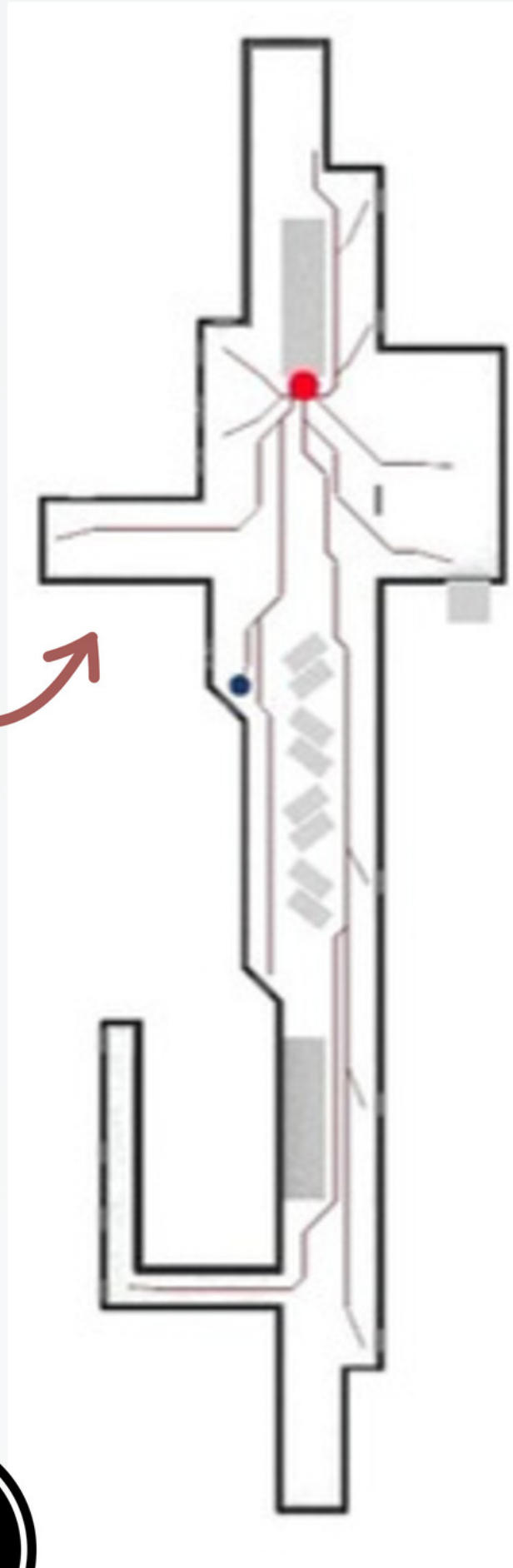
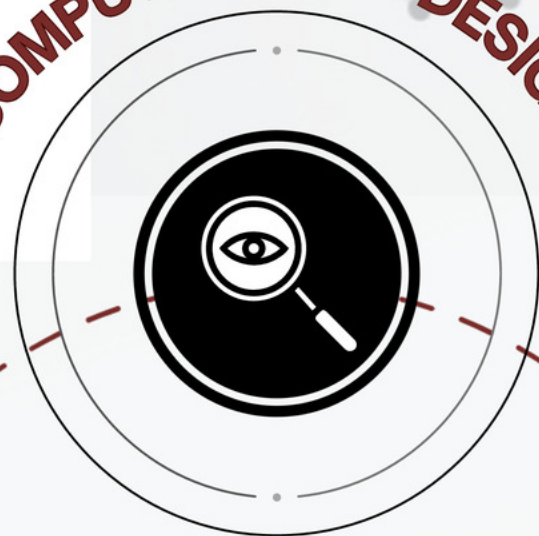
13. Output the position and tangent vectors.



PLAN & PATH

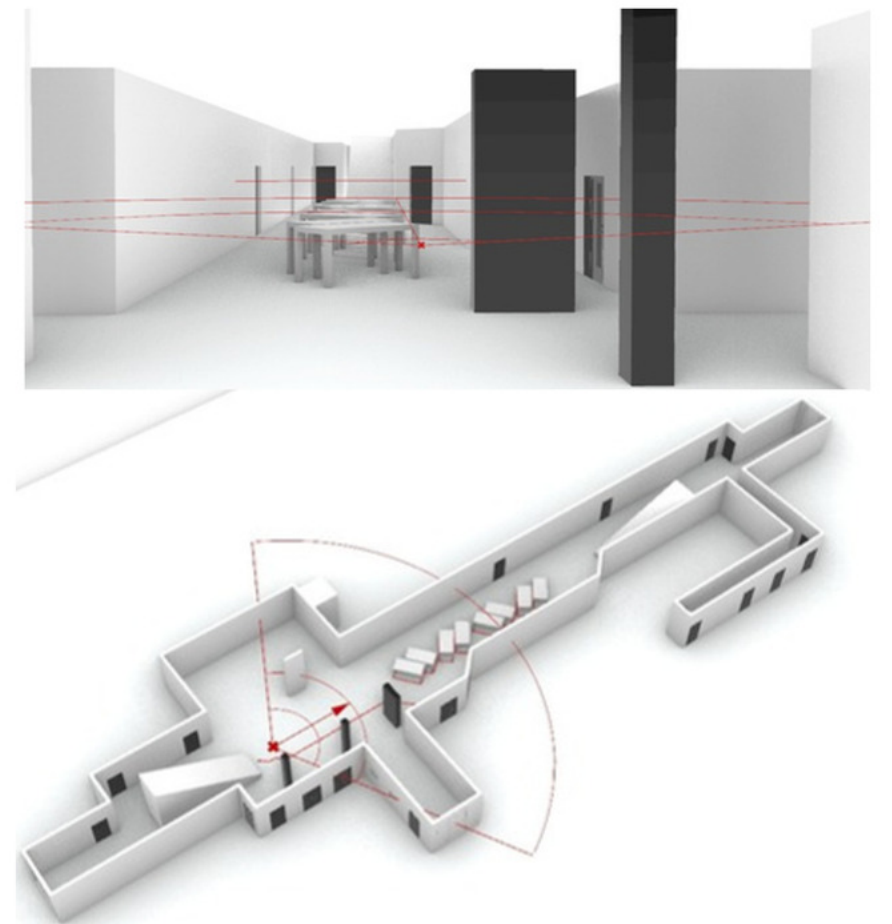
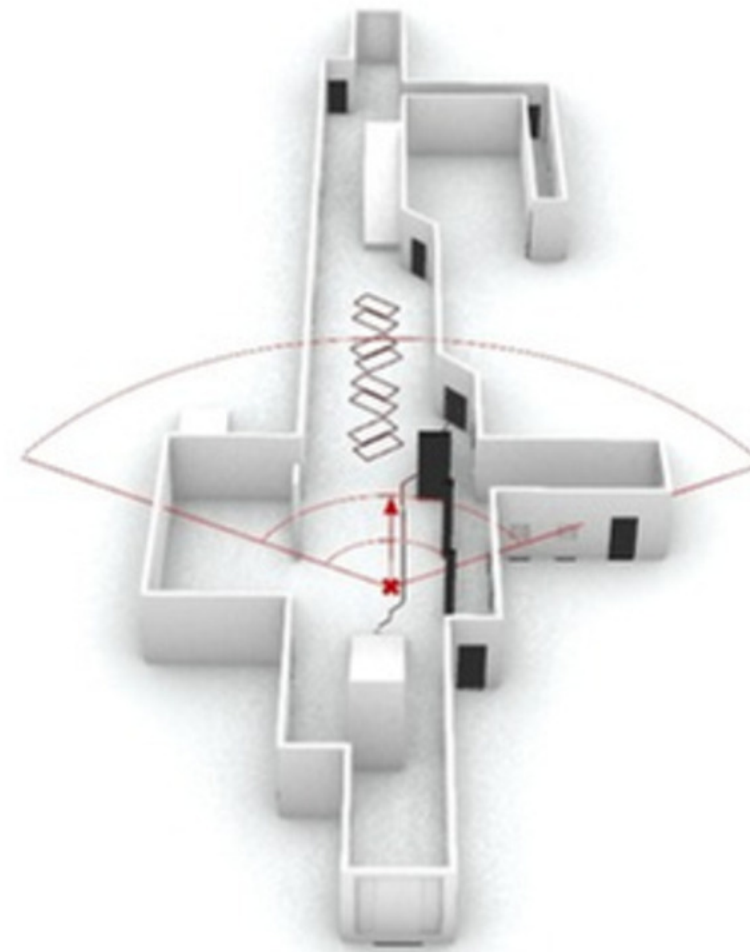


COMPUTATIONAL DESIGN





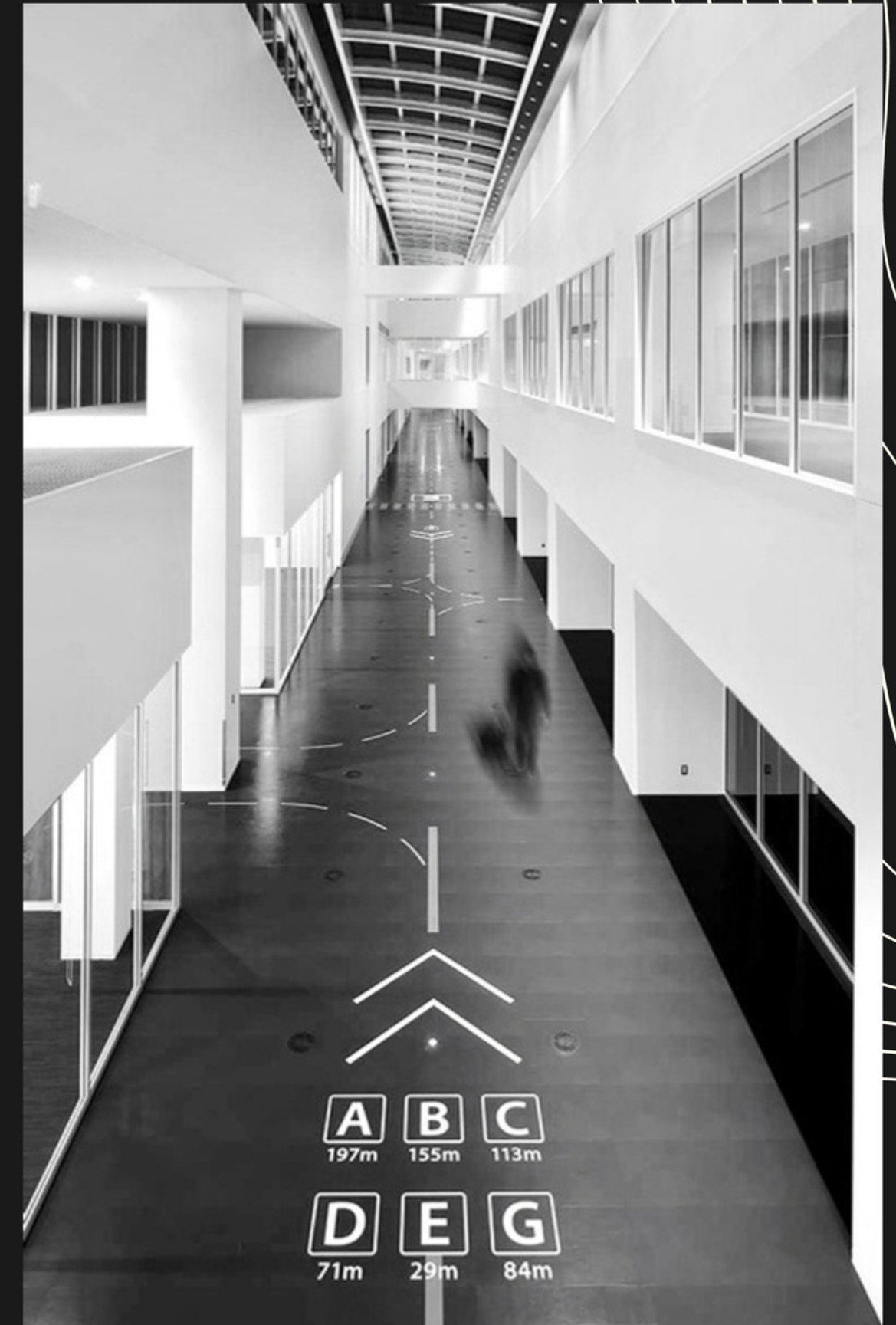
HEAT MAP



THE RESULT OF THE PROJECT

After simulating and analyzing the data in this project, the following results were obtained:

1. The range of the user's readability in the indoor space of the Faculty of Architecture at Iran University of Science and Technology was determined, and in this way, the installation location of the signboard was considered for better navigation at the entrance of the user with a distance of 35 cm.
2. The shortest path to classrooms and conference room was determined by A* algorithm, and paths were drawn for optimum way on the floor that a user use.
3. In the end, in the proposed design(plan) mentioned that the combination of the shortest paths and the guide board should be used, so that the agent can be guided to his desired destination at the right time without getting lost and confusion in the space.



VALUES

What is grid drawing

The grid method is a technique used in art that involves dividing an image into a series of smaller, more manageable sections using a grid. A grid is typically drawn over the reference image, and then students use the grid to draw the image on their own piece of paper, carefully replicating each section of the grid.



The usage of computational design in real life depends on accurate simulation of information

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FUTURE PHASE

In the future, to complete the project, the following items will also be considered:

-Consider the height

- A range of height of each agent in 3D modeling
- The height(position) of sign board for each space

1

-Common path with multiple goals

- Considering various personas in this space such as how to help blind people to navigate

3

-Consider time

- in which way an agent can find his path and target in a quick way without confusion and wasting time

2



MENTORS



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Zahra solbi
parto rahimi
Ali zarrin
Armin kazimi
Mohammad alinia





THE END

